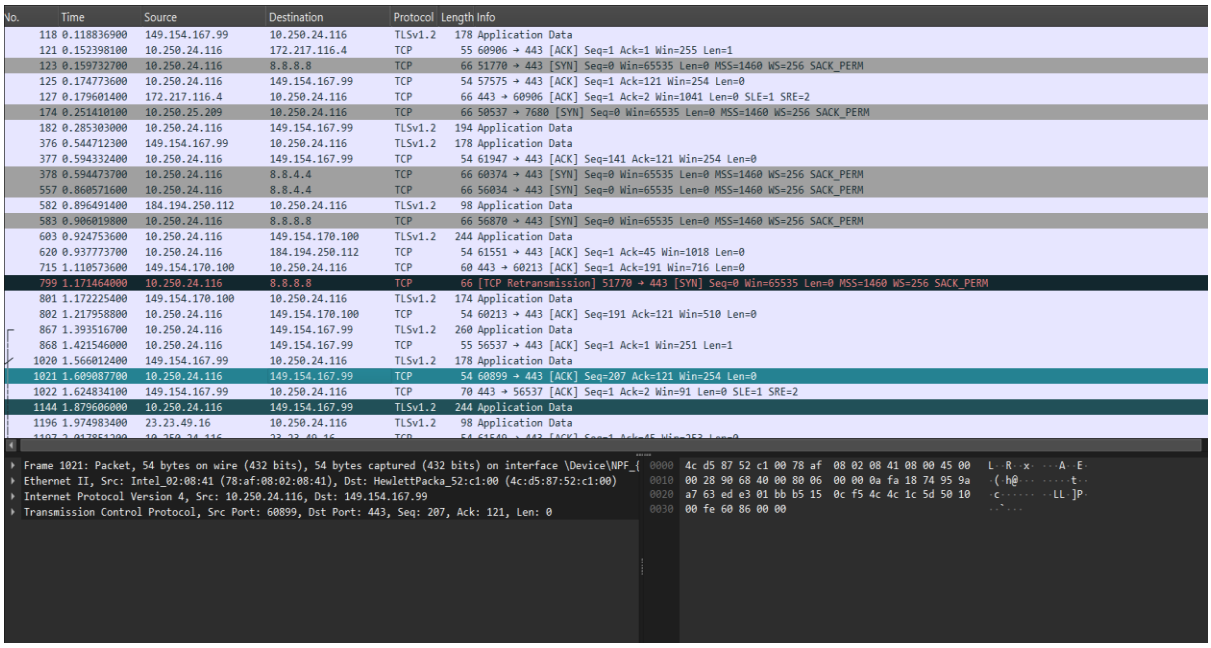
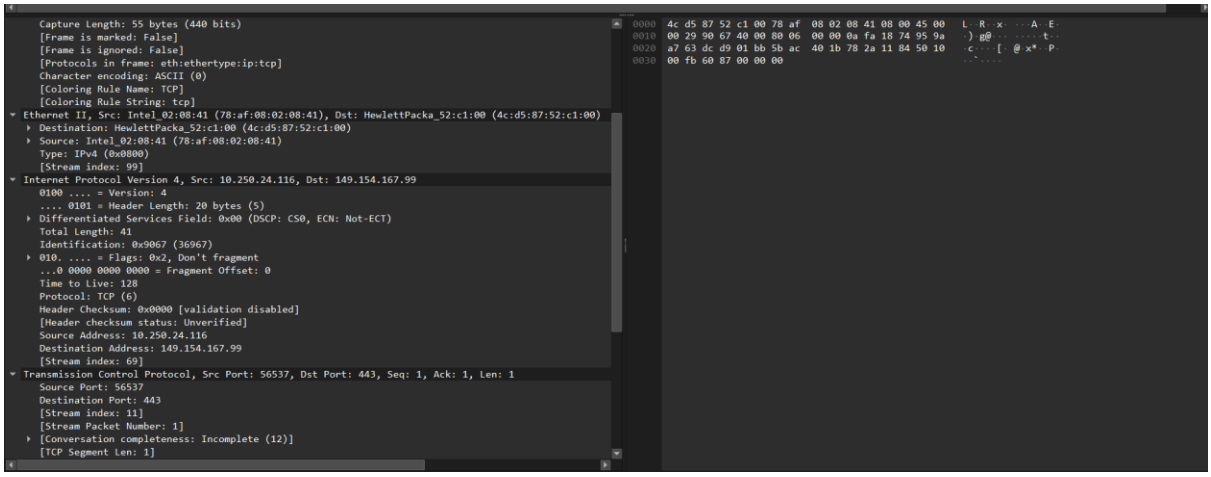
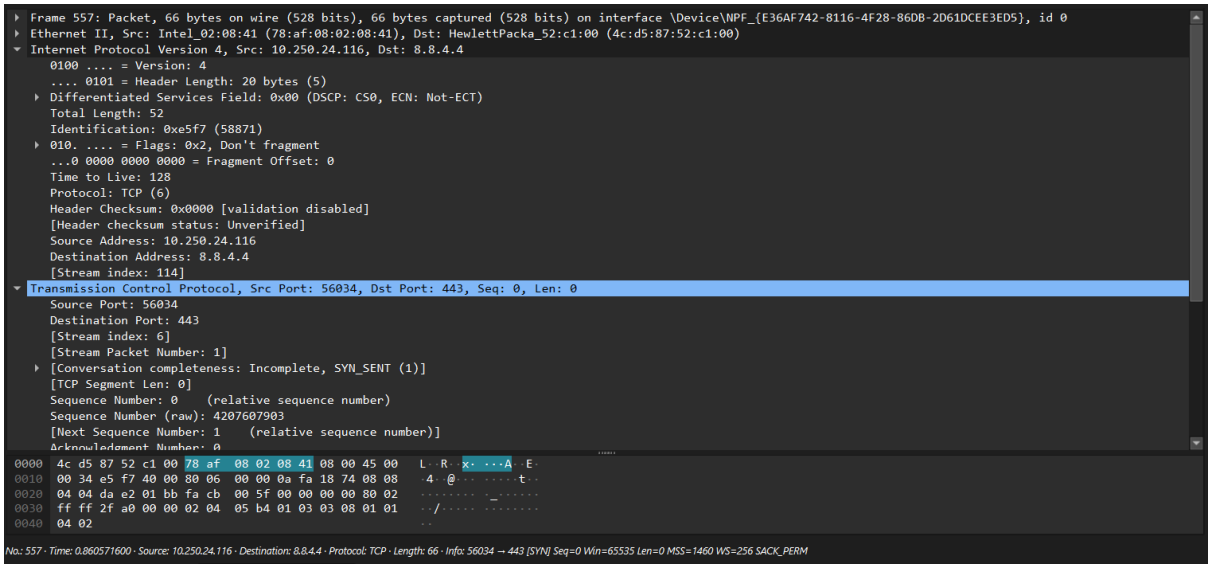
WEEK 1:

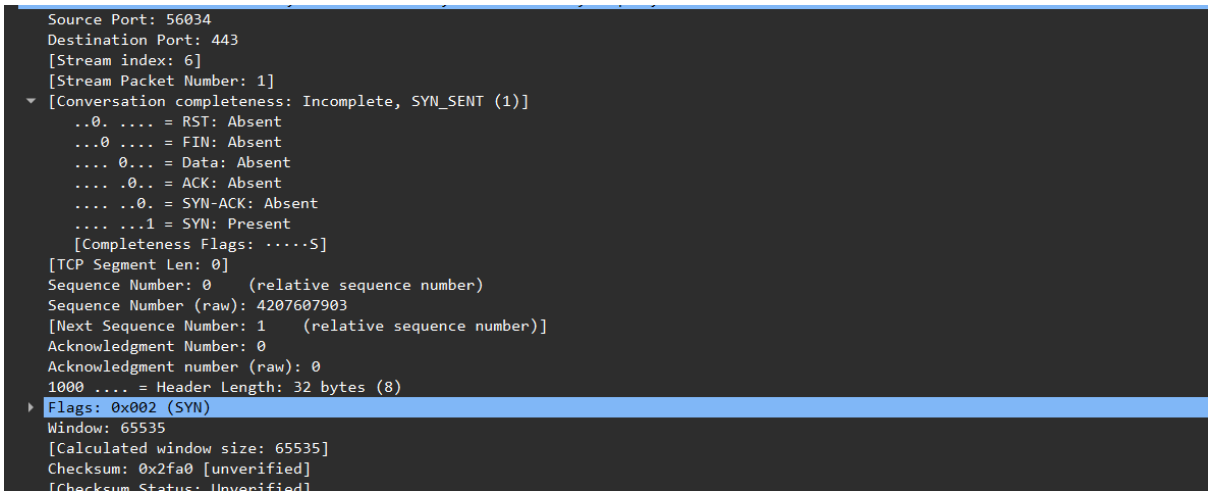
1)TCP FILTER:



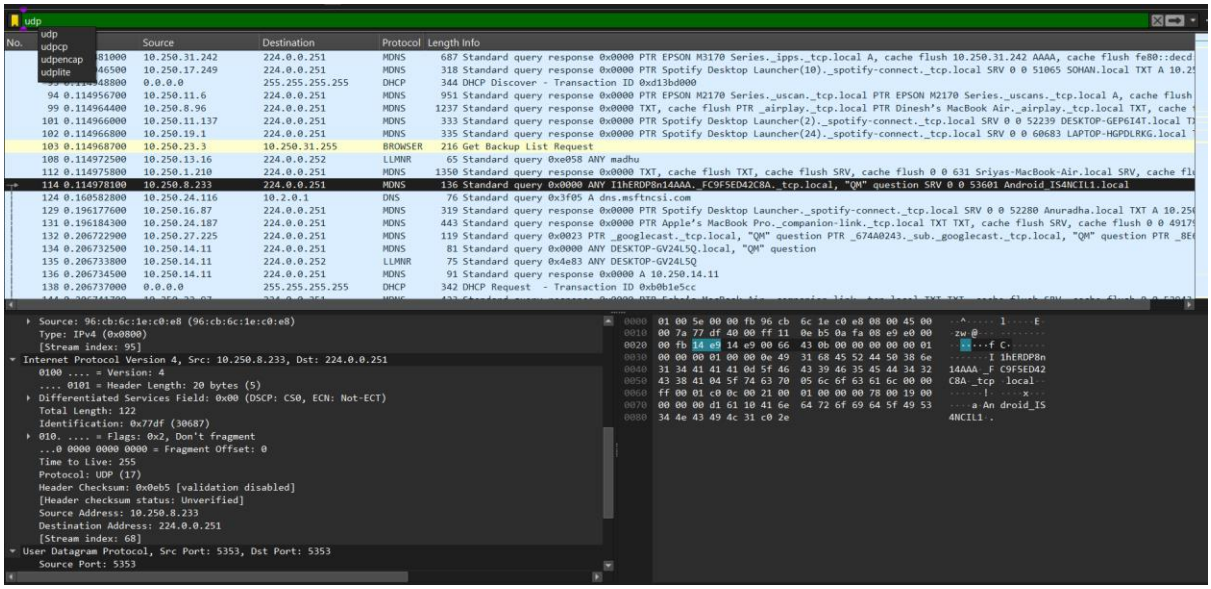



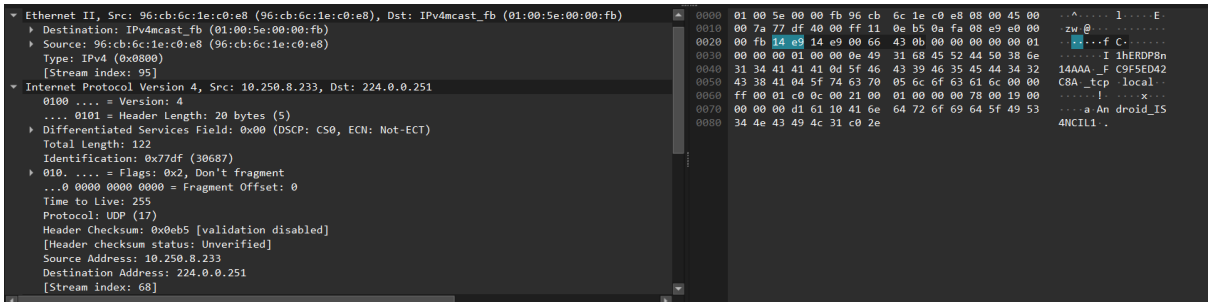

TCP header inspection:

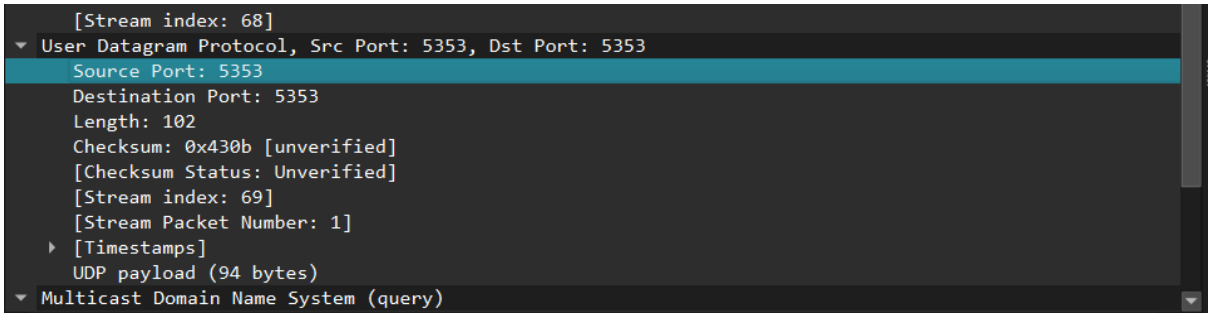

Frame 557: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{E36AF742-8116-4F28-86DB-2D61DCEE3ED5}, id 0
Ethernet II, Src: Intel_02:08:41 (78:af:08:02:08:41), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Internet Protocol Version 4, Src: 10.250.24.116, Dst: 8.8.4.4
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 52
Identification: 0xe5f7 (58871)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.24.116
Destination Address: 8.8.4.4
[Stream index: 114]
Transmission Control Protocol, Src Port: 56034, Dst Port: 443, Seq: 0, Len: 0
Source Port: 56034
Destination Port: 443
[Stream index: 6]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete, SYN_SENT (1)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 4207607903
[Next Sequence Number: 1 (relative sequence number)]
0000 4c d5 87 52 c1 00 78 af 08 02 08 41 08 00 45 00
0010 00 34 e5 f7 40 00 80 06 00 00 0a fa 18 74 08 08
0020 04 04 da e2 01 bb fa cb 00 5f 00 00 00 00 80 02
0030 ff ff 2f a0 00 00 02 04 05 b4 01 03 03 08 01 01
0040 04 02
No.: 557 · Time: 0.860571600 · Source: 10.250.24.116 · Destination: 8.8.4.4 · Protocol: TCP · Length: 66 · Info: 56034 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM


Source Port: 56034
Destination Port: 443
[Stream index: 6]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete, SYN_SENT (1)]
..0. .... = RST: Absent
...0 .... = FIN: Absent
.... 0... = Data: Absent
.... .0.. = ACK: Absent
.... ..0. = SYN-ACK: Absent
.... ...1 = SYN: Present
[Completeness Flags: ·····S]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 4207607903
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1000 .... = Header Length: 32 bytes (8)
Flags: 0x002 (SYN)
Window: 65535
[Calculated window size: 65535]
Checksum: 0x2fa0 [unverified]

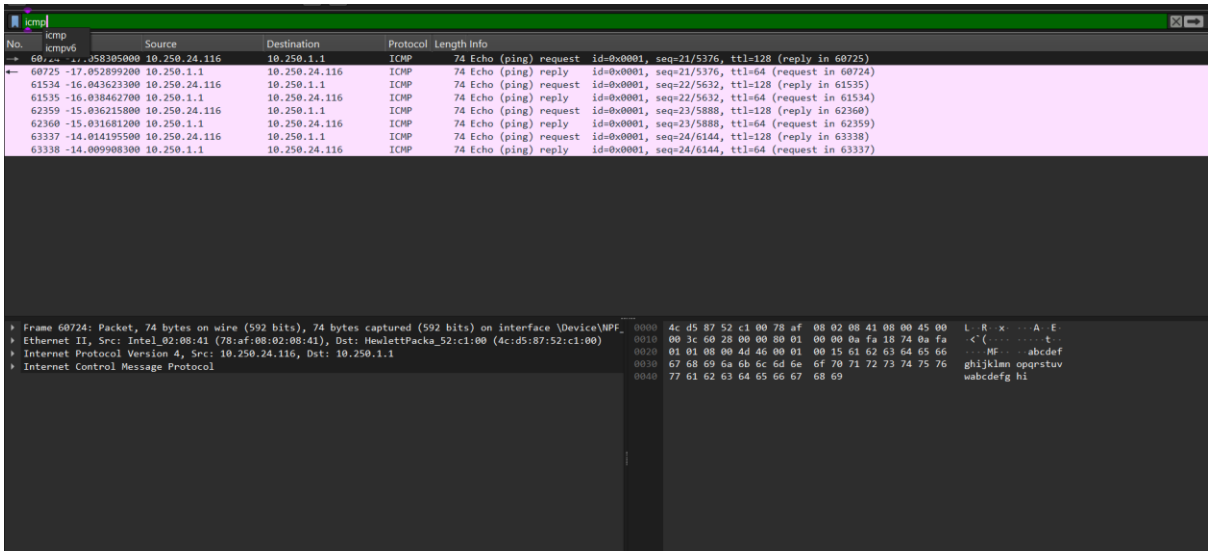
2)UDP :

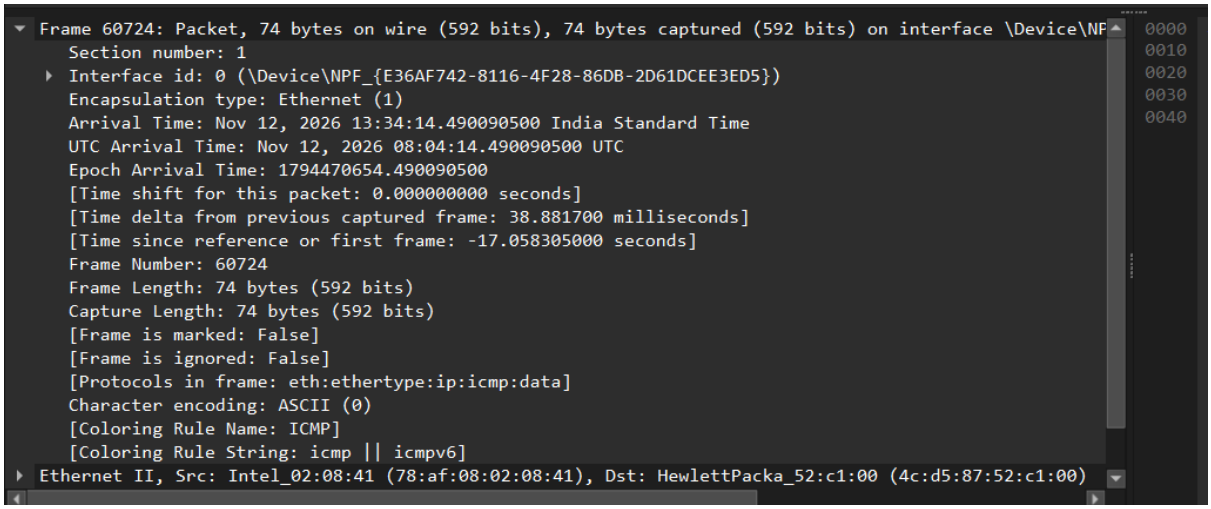

udp
udp
udpcp
udpencap
udplite
No. Source Destination Protocol Length Info
Source: 96:cb:6c:1e:c0:e8 (96:cb:6c:1e:c0:e8)
Type: IPv4 (0x0800)
[Stream index: 95]
Internet Protocol Version 4, Src: 10.250.8.233, Dst: 224.0.0.251
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 122
Identification: 0x77df (30687)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 255
Protocol: UDP (17)
Header Checksum: 0x0eb5 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.8.233
Destination Address: 224.0.0.251
[Stream index: 68]
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Source Port: 5353

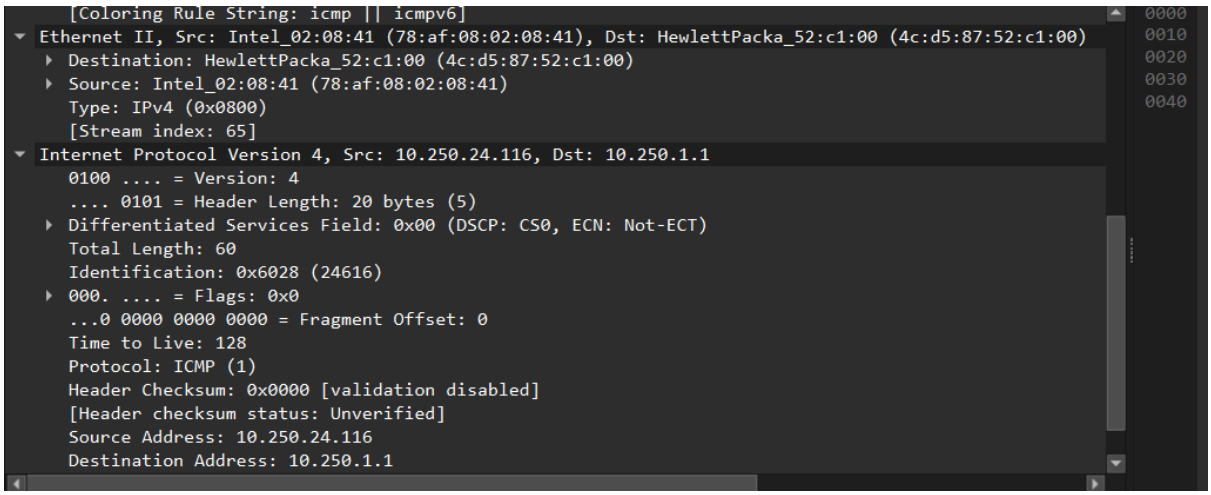

Ethernet II, Src: 96:cb:6c:1e:c0:e8 (96:cb:6c:1e:c0:e8), Dst: IPv4mcast_fb (01:00:5e:00:00:fb)
Destination: IPv4mcast_fb (01:00:5e:00:00:fb)
Source: 96:cb:6c:1e:c0:e8 (96:cb:6c:1e:c0:e8)
Type: IPv4 (0x0800)
[Stream index: 95]
Internet Protocol Version 4, Src: 10.250.8.233, Dst: 224.0.0.251
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 122
Identification: 0x77df (30687)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 255
Protocol: UDP (17)
Header Checksum: 0x0eb5 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.8.233
Destination Address: 224.0.0.251
[Stream index: 68]

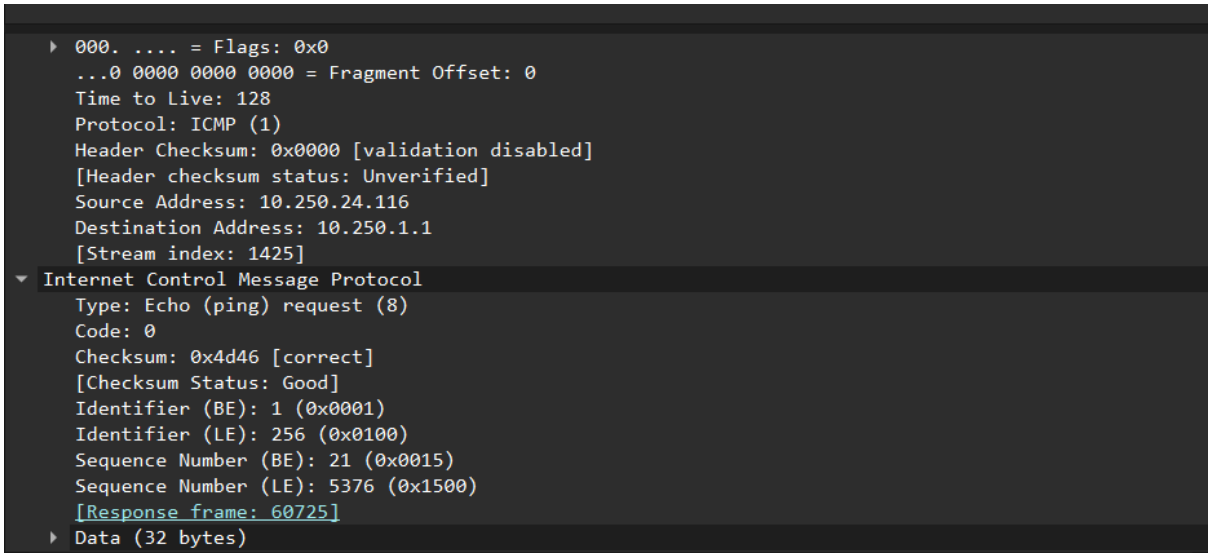

[Stream index: 68]
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Source Port: 5353
Destination Port: 5353
Length: 102
Checksum: 0x430b [unverified]
[Checksum Status: Unverified]
[Stream index: 69]
[Stream Packet Number: 1]
[Timestamps]
UDP payload (94 bytes)
Multicast Domain Name System (query)

ICMP FILTER:


icmp
icmp
icmpv6
No. Source Destination Protocol Length Info
60725 -17.052899200 10.250.1.1 10.250.24.116 ICMP 74 Echo (ping) reply id=0x0001, seq=21/5376, ttl=64 (request in 60724)
61534 -16.043623300 10.250.24.116 10.250.1.1 ICMP 74 Echo (ping) request id=0x0001, seq=22/5632, ttl=128 (reply in 61535)
61535 -16.038462700 10.250.1.1 10.250.24.116 ICMP 74 Echo (ping) reply id=0x0001, seq=22/5632, ttl=64 (request in 61534)
62359 -15.036215800 10.250.24.116 10.250.1.1 ICMP 74 Echo (ping) request id=0x0001, seq=23/5888, ttl=128 (reply in 62360)
62360 -15.031681200 10.250.1.1 10.250.24.116 ICMP 74 Echo (ping) reply id=0x0001, seq=23/5888, ttl=64 (request in 62359)
63337 -14.014195500 10.250.24.116 10.250.1.1 ICMP 74 Echo (ping) request id=0x0001, seq=24/6144, ttl=128 (reply in 63338)
63338 -14.009908300 10.250.1.1 10.250.24.116 ICMP 74 Echo (ping) reply id=0x0001, seq=24/6144, ttl=64 (request in 63337)
Frame 60724: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_
Ethernet II, Src: Intel_02:08:41 (78:af:08:02:08:41), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Internet Protocol Version 4, Src: 10.250.24.116, Dst: 10.250.1.1
Internet Control Message Protocol
0000 4c d5 87 52 c1 00 78 af 08 02 08 41 08 00 45 00
0010 00 3c 60 28 00 00 80 01 00 00 0a fa 18 74 0a fa
0020 01 01 08 00 4d 46 00 01 00 15 61 62 63 64 65 66
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76
0040 77 61 62 63 64 65 66 67 68 69
ghijklmn opqrstuv
wabcdefg hi


Frame 60724: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NP
Section number: 1
Interface id: 0 (\Device\NPF_{E36AF742-8116-4F28-86DB-2D61DCEE3ED5})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 12, 2026 13:34:14.490090500 India Standard Time
UTC Arrival Time: Nov 12, 2026 08:04:14.490090500 UTC
Epoch Arrival Time: 1794470654.490090500
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 38.881700 milliseconds]
[Time since reference or first frame: -17.058305000 seconds]
Frame Number: 60724
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:icmp:data]
Character encoding: ASCII (0)
[Coloring Rule Name: ICMP]
[Coloring Rule String: icmp || icmpv6]
Ethernet II, Src: Intel_02:08:41 (78:af:08:02:08:41), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
0000
0010
0020
0030
0040


[Coloring Rule String: icmp || icmpv6]
Ethernet II, Src: Intel_02:08:41 (78:af:08:02:08:41), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Destination: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Source: Intel_02:08:41 (78:af:08:02:08:41)
Type: IPv4 (0x0800)
[Stream index: 65]
Internet Protocol Version 4, Src: 10.250.24.116, Dst: 10.250.1.1
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 60
Identification: 0x6028 (24616)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: ICMP (1)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.24.116
Destination Address: 10.250.1.1
0000
0010
0020
0030
0040


000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: ICMP (1)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.24.116
Destination Address: 10.250.1.1
[Stream index: 1425]
Internet Control Message Protocol
Type: Echo (ping) request (8)
Code: 0
Checksum: 0x4d46 [correct]
[Checksum Status: Good]
Identifier (BE): 1 (0x0001)
Identifier (LE): 256 (0x0100)
Sequence Number (BE): 21 (0x0015)
Sequence Number (LE): 5376 (0x1500)
[Response frame: 60725]
Data (32 bytes)